

**2025 WINTER SEMESTER**  
**Programming for Data Science Lab (MACSE502) Record**  
**Exercise Program**

Date: 09/07/2026

Week No.: 1

Program S. No. 2

| Title                          | Data Manipulation and Analysis using the Starwars Dataset in R                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                        |                  |                                                        |  |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------------------|--------------------------------------------------------|--|
| <b>Problem Statement</b>       | An AI company maintains model performance records containing: <ul style="list-style-type: none"> <li>• ModelID</li> <li>• Accuracy</li> <li>• Algorithm</li> </ul> Perform: <ol style="list-style-type: none"> <li>1. Select ModelID and Accuracy.</li> <li>2. Filter models with Accuracy greater than 90%.</li> <li>3. Add Status with default value "Production Ready".</li> <li>4. Sort models by Accuracy.</li> <li>5. Calculate average model accuracy.</li> </ol>                                                                                      |                        |                  |                                                        |  |
| <b>Objectives</b>              | <ul style="list-style-type: none"> <li>• Understand how AI model performance data is stored and represented in a Pandas DataFrame.</li> <li>• Filter machine learning models based on conditions, such as accuracy greater than 90%.</li> <li>• Learn how to select specific columns such as ModelID and Accuracy.</li> <li>• Sort AI models according to their accuracy values.</li> <li>• Calculate statistical values such as average, minimum, and maximum model accuracy.</li> <li>• Rank machine learning models based on their performance.</li> </ul> |                        |                  |                                                        |  |
| <b>Dataset URL Link / File</b> | ModelID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Algorithm              | PreviousAccuracy | Accuracy                                               |  |
|                                | M101                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Logistic Regression    | 84.5             | 87.2                                                   |  |
|                                | M102                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Decision Tree          | 87.0             | 89.5                                                   |  |
|                                | M103                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Random Forest          | 90.2             | 93.8                                                   |  |
|                                | M104                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Support Vector Machine | 88.5             | 91.4                                                   |  |
|                                | M105                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | K-Nearest Neighbors    | 82.5             | 85.6                                                   |  |
|                                | M106                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Random Forest          | 91.0             | 94.5                                                   |  |
|                                | M107                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Decision Tree          | 85.5             | 88.3                                                   |  |
|                                | M108                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Neural Network         | 92.0             | 96.1                                                   |  |
|                                | M109                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Logistic Regression    | 86.0             | 90.8                                                   |  |
|                                | M110                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Neural Network         | 93.1             | 95.4                                                   |  |
|                                | M111                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Gradient Boosting      | 89.4             | 92.7                                                   |  |
|                                | M112                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Gradient Boosting      | 91.8             | 94.1                                                   |  |
| <b>Functions Details</b>       | Sl. No.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Function               | Package          | Description of function                                |  |
|                                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | pd.DataFrame()         | Pandas           | Creates a DataFrame from the AI model data dictionary. |  |
|                                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | pd.read_csv()          | Pandas           | Reads AI model records from a CSV file.                |  |
|                                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | dropna()               | Pandas           | Removes rows containing missing or invalid values      |  |
|                                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | drop_duplicates()      | Pandas           | Removes duplicate records based on ModelID.            |  |

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|                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                     |        |                                                                   |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--------|-------------------------------------------------------------------|
|                                                                                   | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | sort_values()       | Pandas | Sorts the models according to their accuracy values.              |
|                                                                                   | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | mean()              | Pandas | Calculates the average accuracy of all AI models.                 |
|                                                                                   | 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | groupby() and agg() | Pandas | Groups models by algorithm and calculates performance statistics. |
|                                                                                   | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | to_csv()            | Pandas | Saves the analysed model results into a CSV file.                 |
| <b>Steps/<br/>Process/<br/>Procedure/<br/>Algorithm/<br/>Flowchart/<br/>Model</b> | <ol style="list-style-type: none"> <li>1. Import the Pandas package.</li> <li>2. Create the AI model dataset containing ModelID, Algorithm, PreviousAccuracy and Accuracy.</li> <li>3. Convert the dataset into a Pandas DataFrame.</li> <li>4. Clean the dataset by removing missing values, duplicate ModelIDs and invalid accuracy values.</li> <li>5. Select only the ModelID and Accuracy columns.</li> <li>6. Filter the models whose Accuracy is greater than 90%.</li> <li>7. Add a Status column with the default value "Production Ready".</li> <li>8. Change the status to "Needs Improvement" for models having Accuracy below 90%.</li> <li>9. Sort the models in descending order based on Accuracy.</li> <li>10. Assign ranks to the models, where the highest-accuracy model receives Rank 1.</li> <li>11. Calculate the improvement of each model using:<br/>Improvement = Accuracy – PreviousAccuracy</li> <li>12. Calculate the average accuracy of all models.</li> <li>13. Group the models by Algorithm and calculate minimum, maximum and average accuracy.</li> <li>14. Identify the best-performing, lowest-performing and most-improved models.</li> <li>15. Display all the analysis results in the console.</li> <li>16. Generate an automated model evaluation report.</li> <li>17. Save the analysed results in a CSV file and the report in a text file.</li> </ol> |                     |        |                                                                   |
| <b>Program</b>                                                                    | <pre>import sys from pathlib import Path  import pandas as pd  from analysis_core import (     analyse_models,     clean_data,     create_algorithm_summary,     create_report,     create_sample_data,     perform_basic_operations )</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                     |        |                                                                   |

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```
#
=====

# HEADING FUNCTION
# This keeps console output neat and readable.
#
=====

def print_heading(title):
    print("\n" + "=" * 72)
    print(title)
    print("=" * 72)

#
=====

# MAIN CONSOLE PROGRAM
#
# Run with sample data:
# python console_app.py
#
# Run with CSV data:
# python console_app.py sample_ai_models.csv
#
=====

def main():
    print_heading("AI MODEL PERFORMANCE ANALYSIS -
    CONSOLE MODE")

    csv_file = None

    # If the user gives a file path, use that CSV.
    if len(sys.argv) > 1:
        csv_file = sys.argv[1]

    if csv_file:
        csv_path = Path(csv_file)

        if not csv_path.exists():
            print(f"Error: File not found: {csv_path}")
            return

        print(f"Using CSV file: {csv_path}")
        raw_data = pd.read_csv(csv_path)

    else:
        print("Using built-in sample data.")
```

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```
raw_data = create_sample_data()

try:
    cleaned_data = clean_data(raw_data)
except ValueError as error:
    print(f'Data error: {error}')
    return

analysed_data = analyse_models(
    cleaned_data,
    production_threshold=90.0
)

results = perform_basic_operations(
    analysed_data
)

print_heading("ORIGINAL CLEANED DATA")
print(
    cleaned_data.to_string(index=False)
)

print_heading("1. SELECT MODELID AND ACCURACY")
print(
    results["selected_columns"]
    .to_string(index=False)
)

print_heading(
    "2. FILTER MODELS WITH ACCURACY GREATER THAN 90%"
)
print(
    results["models_above_90"][
        ["ModelID", "Algorithm", "Accuracy"]
    ].to_string(index=False)
)

print_heading(
    "3. ADD STATUS WITH DEFAULT 'PRODUCTION READY'"
)
print(
    results["status_table"]
    .to_string(index=False)
)

print_heading("4. SORT MODELS BY ACCURACY")
print(
```

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```
results["sorted_models"][
    ["ModelID", "Algorithm", "Accuracy"]
].to_string(index=False)
)

print_heading("5. CALCULATE AVERAGE MODEL
ACCURACY")
print(
    f'Average model accuracy: '
    f'{results["average_accuracy"]:.2f}%'
)

print_heading("MODEL RANKING")
print(
    analysed_data[
        [
            "Rank",
            "ModelID",
            "Algorithm",
            "PreviousAccuracy",
            "Accuracy",
            "Improvement",
            "Status"
        ]
    ].to_string(index=False)
)

print_heading("ALGORITHM PERFORMANCE COMPARISON")
algorithm_summary = create_algorithm_summary(
    analysed_data
)

print(
    algorithm_summary.to_string(index=False)
)

print_heading("ACCURACY IMPROVEMENT TRACKING")
print(
    analysed_data[
        [
            "ModelID",
            "PreviousAccuracy",
            "Accuracy",
            "Improvement",
            "ImprovementStatus"
        ]
    ].to_string(index=False)
)
```

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```
report = create_report(
    analysed_data,
    production_threshold=90.0
)

print_heading("AUTOMATED EVALUATION REPORT")
print(report)

# Save analysed data.
analysed_data.to_csv(
    "console_ai_model_results.csv",
    index=False
)

# Save the report.
with open(
    "console_ai_model_report.txt",
    "w",
    encoding="utf-8"
) as report_file:
    report_file.write(report)

print_heading("FILES CREATED")
print("console_ai_model_results.csv")
print("console_ai_model_report.txt")

if __name__ == "__main__":
    main()
```

**Output  
Snapshots  
With  
Explanation/  
Observations**

```
ORIGINAL CLEANED DATA
=====
ModelID      Algorithm  PreviousAccuracy  Accuracy
M101  Logistic Regression      84.5      87.2
M102  Decision Tree           87.0      89.5
M103  Random Forest           90.2      93.8
M104  Support Vector Machine    88.5      91.4
M105  K-Nearest Neighbors        82.5      85.6
M106  Random Forest           91.0      94.5
M107  Decision Tree           85.5      88.3
M108  Neural Network           92.0      96.1
M109  Logistic Regression      86.0      90.8
M110  Neural Network           93.1      95.4
M111  Gradient Boosting        89.4      92.7
M112  Gradient Boosting        91.8      94.1
```

Figure 2.1 Original Cleaned Dataset

The ModelID and Accuracy columns were selected successfully. Models having accuracy greater than 90% were filtered from the dataset.

```
1. SELECT MODELID AND ACCURACY
=====
ModelID  Accuracy
M108     96.1
M110     95.4
M106     94.5
M112     94.1
M103     93.8
M111     92.7
M104     91.4
M109     90.8
M102     89.5
M107     88.3
M101     87.2
M105     85.6
```

Figure 1.2 Selected ModelID and Accuracy

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2. FILTER MODELS WITH ACCURACY GREATER THAN 90%

| ModelID | Algorithm              | Accuracy |
|---------|------------------------|----------|
| M108    | Neural Network         | 96.1     |
| M110    | Neural Network         | 95.4     |
| M106    | Random Forest          | 94.5     |
| M112    | Gradient Boosting      | 94.1     |
| M103    | Random Forest          | 93.8     |
| M111    | Gradient Boosting      | 92.7     |
| M104    | Support Vector Machine | 91.4     |
| M109    | Logistic Regression    | 90.8     |

Figure 1.3 Models Accuracy Greater than 90%

A Status column was added. Models with accuracy greater than or equal to 90% were marked as “Production Ready”, while the remaining models were marked as “Needs Improvement”.

3. ADD STATUS WITH DEFAULT "PRODUCTION READY"

| ModelID | Accuracy | Status            |
|---------|----------|-------------------|
| M108    | 96.1     | Production Ready  |
| M110    | 95.4     | Production Ready  |
| M106    | 94.5     | Production Ready  |
| M112    | 94.1     | Production Ready  |
| M103    | 93.8     | Production Ready  |
| M111    | 92.7     | Production Ready  |
| M104    | 91.4     | Production Ready  |
| M109    | 90.8     | Production Ready  |
| M102    | 89.5     | Needs Improvement |
| M107    | 88.3     | Needs Improvement |
| M101    | 87.2     | Needs Improvement |
| M105    | 85.6     | Needs Improvement |

Figure 1.4 Status Column and Models sorted by accuracy

5. CALCULATE AVERAGE MODEL ACCURACY

Average model accuracy: 91.62%

MODEL RANKING

| Rank | ModelID | Algorithm              | PreviousAccuracy | Accuracy | Improvement | Status            |
|------|---------|------------------------|------------------|----------|-------------|-------------------|
| 1    | M108    | Neural Network         | 92.0             | 96.1     | 4.1         | Production Ready  |
| 2    | M110    | Neural Network         | 93.1             | 95.4     | 2.3         | Production Ready  |
| 3    | M106    | Random Forest          | 91.0             | 94.5     | 3.5         | Production Ready  |
| 4    | M112    | Gradient Boosting      | 91.8             | 94.1     | 2.3         | Production Ready  |
| 5    | M103    | Random Forest          | 90.2             | 93.8     | 3.6         | Production Ready  |
| 6    | M111    | Gradient Boosting      | 89.4             | 92.7     | 3.3         | Production Ready  |
| 7    | M104    | Support Vector Machine | 88.5             | 91.4     | 2.9         | Production Ready  |
| 8    | M109    | Logistic Regression    | 86.0             | 90.8     | 4.8         | Production Ready  |
| 9    | M102    | Decision Tree          | 87.0             | 89.5     | 2.5         | Needs Improvement |
| 10   | M107    | Decision Tree          | 85.5             | 88.3     | 2.8         | Needs Improvement |
| 11   | M101    | Logistic Regression    | 84.5             | 87.2     | 2.7         | Needs Improvement |
| 12   | M105    | K-Nearest Neighbors    | 82.5             | 85.6     | 3.1         | Needs Improvement |

Figure 1.5 Average Model Accuracy

The average accuracy of all models was calculated. Accuracy improvement was also calculated by subtracting PreviousAccuracy from the current Accuracy.

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ALGORITHM PERFORMANCE COMPARISON

=====

| Algorithm              | ModelCount | MinimumAccuracy | MaximumAccuracy | AverageAccuracy | AverageImprovement |
|------------------------|------------|-----------------|-----------------|-----------------|--------------------|
| Neural Network         | 2          | 95.4            | 96.1            | 95.75           | 3.20               |
| Random Forest          | 2          | 93.8            | 94.5            | 94.15           | 3.55               |
| Gradient Boosting      | 2          | 92.7            | 94.1            | 93.40           | 2.80               |
| Support Vector Machine | 1          | 91.4            | 91.4            | 91.40           | 2.90               |
| Logistic Regression    | 2          | 87.2            | 90.8            | 89.00           | 3.75               |
| Decision Tree          | 2          | 88.3            | 89.5            | 88.90           | 2.65               |
| K-Nearest Neighbors    | 1          | 85.6            | 85.6            | 85.60           | 3.10               |

Figure 1.6 Models Grouped according to their algorithm and other metrics

The models were grouped according to their algorithms, and minimum, maximum and average accuracy values were calculated for each algorithm.

```
SUMMARY
Total models analysed: 12
Production threshold: 90.0%
Average model accuracy: 91.62%
Production-ready models: 8
Models needing improvement: 4

BEST MODEL
Model ID: M108
Algorithm: Neural Network
Accuracy: 96.10%

LOWEST-PERFORMING MODEL
Model ID: M105
Algorithm: K-Nearest Neighbors
Accuracy: 85.60%

MOST IMPROVED MODEL
Model ID: M109
Algorithm: Logistic Regression
Improvement: 4.80 percentage points

BEST ALGORITHM
Algorithm: Neural Network
Average accuracy: 95.75%
```

The program identified the best-performing model, lowest-performing model, most-improved model and best-performing algorithm.

Figure 1.7 Final Output

### Conclusions

The exercise successfully demonstrated how Pandas can be used to analyse AI model performance data. The program selected, filtered, sorted and ranked the models, calculated average accuracy, compared algorithm performance and tracked accuracy improvement. It also generated an automated evaluation report and saved the final results in CSV and text formats.

### References

1. Michael Freeman and Joel Ross, Programming Skills for Data Science: Start Writing Code to Wrangle, Analyze, and Visualize Data with R, Addison-Wesley, 2018
2. Python Program ([https://github.com/hasnainmakada-99/Data-Science-Codes/blob/main/WEEK%201/Excercise Program Console.py](https://github.com/hasnainmakada-99/Data-Science-Codes/blob/main/WEEK%201/Excercise%20Program%20Console.py))
3. Class lecture link ([https://drive.google.com/file/d/15diEukCtrPLimox63oa9ce1Ut4qVMUy/view?usp=drive link](https://drive.google.com/file/d/15diEukCtrPLimox63oa9ce1Ut4qVMUy/view?usp=drive_link))